AGHA MALIK BAIG MIRZA

Computer Engineer
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SUMMARY

M.S. Computer Engineering student at CSUN, graduating December 2026, seeking full-time RTL design, design verification, ASIC/SoC and FPGA roles. Built UVM testbenches for AMBA APB and AXI4 memory slaves in coursework and personal projects, with constrained-random stimulus and SVA assertions on the AXI4 environment. Strong in Verilog and hands-on with SystemVerilog.

TECHNICAL SKILLS

- **Verification:** UVM, SVA, constrained-random verification, functional coverage, directed testbenches.
- **HDLs and programming:** Verilog, SystemVerilog, RTL design and debugging, Python, C.
- **Tools:** Linux command line, shell scripts, Synopsys VCS and DVE, QuestaSim/ModelSim, Xilinx Vivado, GTKWave, EDA Playground.
- **Protocols:** AMBA APB, AXI4, SPI, I2C, UART, synchronous and asynchronous FIFO.

ACADEMIC PROJECT

Design and UVM Verification of AXI4 Memory | github.com/mirzamalikbaig/axi4_uvm

- Wrote a UVM environment for a memory-mapped AXI4 slave covering the AW, W, B, AR and R channels.
- Verified VALID/READY handshakes, independent AW/W timing, backpressure, WSTRB, responses, reset behavior and read-after-write data integrity.
- Added scoreboard-based memory checking and functional coverage for transactions and handshakes.
- Ran VCS simulation and coverage from shell scripts: 98.15% functional coverage, 0 UVM errors, 12 directed scenarios.

PERSONAL PROJECTS

Design and UVM Verification of APB Memory | github.com/mirzamalikbaig/apb_uvm

- Designed an AMBA APB memory slave and verified it with a UVM environment: driver, monitor, reference-memory scoreboard and functional coverage.
- Wrote a directed sequence of 14 transfers checking write-then-read data integrity, including boundary values (0x00000000, 0xFFFFFFFF) and alternating patterns.
- Debugged protocol and data mismatches with scoreboard failures and VCS waveforms, fixing RTL and testbench issues before coverage closure.
- Achieved 100% functional, 100% line and branch, 70.43% toggle and 60% condition coverage.

Asynchronous FIFO Design and Verification | github.com/mirzamalikbaig/async_fifo

- Designed a 16-entry, 8-bit asynchronous FIFO with separate write and read clocks, Gray-coded pointers and two-flop synchronizers for clock domain crossing.
- Verified it with a self-checking scoreboard, seven directed tests and six SVA assertions.
- Ran the seven-test regression from a Python script and GitHub Actions using Verilator.

EXPERIENCE

Verification Intern, Aisemi Pvt. Ltd. | Jan 2024 – May 2024

- Five-month internship on SystemVerilog, RTL verification fundamentals and semiconductor concepts.
- Developed and simulated SystemVerilog RTL and testbenches for combinational and sequential logic.
- Ran Synopsys simulations, inspected waveforms and debugged functional mismatches.

Industrial Trainee, Electronics Corporation of India Limited (ECIL) | Nov 2019 – Mar 2020

- Industrial training at ECIL, a Government of India enterprise, with exposure to electronic systems, digital circuits and laboratory practices.
- Performed supervised circuit testing, signal measurement and basic hardware troubleshooting.

EDUCATION

M.S. in Computer Engineering, California State University, Northridge (CSUN) | Aug 2024 – Dec 2026

Coursework: Verilog & SystemVerilog, ASIC Development, System-on-Chip Design, Computer Systems Architecture, Computer Arithmetic Design.

B.E. in Electronics & Communication Engineering, Osmania University | Jul 2021 – Jun 2023

Coursework: Digital Logic Design, VLSI Design, Microprocessors & Microcontrollers, Embedded Systems.